

Dataset	$\#F_{total}$	$\min(Var(F))$	$\max(Var(F))$	$\min(F)$	$\max(F)$
Global	4550	0	8286433	0	7689680
$\hookrightarrow$ Lagniez [?]	1948	5	229100	10	399794
$\hookrightarrow$ Soos [?]	1823	2	8286433	1	7689680
$\hookrightarrow$ Plazar [?]	501	14	486193	31	2598178
$\hookrightarrow$ Sundermann [?]	278	0	31012	0	350120

Table 1: Dataset summary. The first column indicates the dataset, and the $\#F$ column indicates how many formulae the dataset contains. The following columns indicate the minimum and maximum number of variables (resp. clauses) in the dataset.

Dataset	$\#F_{total}$	$\#D4$	$\#\neg D4$	$\#+D4_{ld}$	$\#+CK$	$\#+DivKC$
Global	4550	185	670	1	79	55
$\hookrightarrow$ Lagniez [?]	1948	53	213	1	39	23
$\hookrightarrow$ Soos [?]	1823	100	393	0	40	28
$\hookrightarrow$ Plazar [?]	501	31	61	0	0	4
$\hookrightarrow$ Sundermann [?]	278	1	3	0	0	0

Table 2: Experimental results regarding the scalability of Algorithm ??. Column $\#F_{total}$ indicates the total number of formulae in each dataset. The next column shows the number of formulae compiled only by D4 [?] but not by Algorithm ??. Column $\#\neg D4$ shows the number of formulae not compiled by D4. The last column indicates the number of formulae that were only compiled by Algorithm ??, but not by D4.

Dataset	$\#F_{total}$	$\#\neg D4$	$\#\neg D4 \wedge CapKC$	$\#\neg D4 \wedge DivKC$
Global	4550	670	80	114
$\hookrightarrow$ Lagniez [?]	1948	213	40	52
$\hookrightarrow$ Soos [?]	1823	393	40	58
$\hookrightarrow$ Plazar [?]	501	61	0	4
$\hookrightarrow$ Sundermann [?]	278	3	0	0

Table 3: Experimental results regarding the scalability of Algorithm ??. Column $\#F_{total}$ indicates the total number of formulae in each dataset. The next column shows the number of formulae compiled only by D4 [?] but not by Algorithm ??. Column $\#\neg D4$ shows the number of formulae not compiled by D4. The last column indicates the number of formulae that were only compiled by Algorithm ??, but not by D4.

Dataset	#F	$Y_l \leq R_F $	$Y_h \geq R_F $	Coverage	$ R_{G_P} \leq R_F \leq R_{G_U} $
Global	3670	0.969	0.857	0.826	1.000
$\hookrightarrow$ Lagniez [?]	1689	0.969	0.895	0.864	1.000
$\hookrightarrow$ Soos [?]	1307	0.969	0.906	0.875	1.000
$\hookrightarrow$ Plazar [?]	403	0.968	0.789	0.757	1.000
$\hookrightarrow$ Sundermann [?]	271	0.967	0.487	0.454	1.000

Table 4: Experimental results for Algorithm ???. Column #F indicates with how many formulae the statistics have been computed. The 'Coverage' column indicates how often $|R_F|$ is within the confidence interval $[Y_l; Y_h]$ and thus measures the accuracy of our method.

Dataset	#F	$Y_l \leq R_F $	$Y_h \geq R_F $	Coverage	$ R_F \leq R_{G_U} $
Global	517	0.994	0.986	0.981	1.000
$\hookrightarrow$ Lagniez [?]	230	0.996	0.978	0.974	1.000
$\hookrightarrow$ Soos [?]	199	0.990	0.995	0.985	1.000
$\hookrightarrow$ Plazar [?]	45	1.000	1.000	1.000	1.000
$\hookrightarrow$ Sundermann [?]	43	1.000	0.977	0.977	1.000

Table 5: Experimental results for CapKC. Column #F indicates with how many formulae the statistics have been computed. The 'Coverage' column indicates how often $|R_F|$ is within the confidence interval $[Y_l; Y_h]$ and thus measures the accuracy of our method.

Dataset	#F	$Y_l \geq R_{G_P} $	$Y_h \leq R_{G_U} $	Both	$median(r_c)$	$max(r_c)$
Global	3669	0.807	0.836	0.721	4.17e-9	0.54
$\hookrightarrow$ Lagniez [?]	1689	0.844	0.869	0.760	1.8e-9	0.54
$\hookrightarrow$ Soos [?]	1307	0.920	0.893	0.834	1.75e-7	0.0962
$\hookrightarrow$ Plazar [?]	403	0.697	0.769	0.610	1.64e-13	0.0153
$\hookrightarrow$ Sundermann [?]	270	0.196	0.448	0.093	1.69e-13	0.281

Table 6: Experimental results comparing the bounds obtained with Algorithm ?? and with Lemma ??. Column #F indicates with how many formulae the statistics have been computed. The 'Both' column indicates how often we have $Y_l \geq |R_{G_P}| \wedge Y_l \leq |R_F|$ and $Y_h \leq |R_{G_U}| \wedge Y_h \geq |R_F|$. The last two columns represent the observed median and maximum values of the ratio $r_c = \frac{\min(Y_h, |R_{G_U}|) - \max(Y_l, |R_{G_P}|)}{|R_{G_U}| - |R_{G_P}|}$, which was calculated exclusively if $Y_l \leq |R_F| \leq Y_h$. The number of formulae on which the last two columns are computed can easily be obtained by multiplying the #F column with the 'Coverage' column in Table 4.

Dataset	#F	$Y_h \leq R_{GU} $	$median(r_c)$	$max(r_c)$
Global	517	0.971	0.0118	0.0257
$\hookrightarrow$ Lagniez [?]	230	0.970	0.0122	0.0257
$\hookrightarrow$ Soos [?]	199	0.990	0.0121	0.0257
$\hookrightarrow$ Plazar [?]	45	0.956	0.00864	0.0257
$\hookrightarrow$ Sundermann [?]	43	0.907	0.00143	0.0169

Table 7: CapKC.

Dataset	#F	$l \leq Y_{\text{ApproxMC}} \leq h$	$l \leq Y_{\text{DivKC-AMC}} \leq h$	$l \leq Y_{\text{CapKC-AMC}} \leq h$
Global	2505	0.977	0.901	0.984
$\hookrightarrow$ Lagniez [?]	1211	1.000	0.916	0.982
$\hookrightarrow$ Soos [?]	1104	0.969	0.885	0.984
$\hookrightarrow$ Plazar [?]	169	0.858	0.893	1.000
$\hookrightarrow$ Sundermann [?]	21	1.000	0.952	1.000

Table 8: CapKC.